Suyog Mainali

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PROFESSIONAL SUMMARY

Senior Data Engineer with 7+ years architecting enterprise-scale data platforms, ETL/ELT pipelines, and streaming infrastructure at 10TB+ daily scale on AWS, GCP, and Azure. Production expertise across Databricks, Apache Spark, Airflow, dbt, Kafka, and Snowflake; dimensional modeling, data quality, and SLA-governed delivery from design through support. Deep HIPAA/HITRUST healthcare experience; equally effective in financial, regulatory, and tech domains. Hands-on with LangChain/LangGraph agentic AI, RAG, vector embeddings, and production ML feature pipelines.

CORE TECHNICAL SKILLS

Languages & Query	Python (advanced; pandas, NumPy, PySpark, FastAPI, scikit-learn, XGBoost), SQL (expert; window functions, CTEs, stored procedures, query optimization), Java, Groovy, R, Bash
Pipelines & Orchestration	Apache Airflow (multi-stage workflows, 99.8% uptime), dbt (transformation, testing, docs), Spark (batch & streaming), Kafka, ETL/ELT, lakehouse & medallion architecture
Data Warehousing & Modeling	Snowflake, Redshift, BigQuery, Databricks (Delta Lake, Unity Catalog), DuckDB, PostgreSQL; Kimball dimensional modeling (star schema, fact/dim, SCD1/2), canonical datasets, semantic layers
Cloud & IaC	AWS (S3, EMR, Glue, Lambda, Step Functions, Redshift, CDK), GCP (BigQuery, Dataflow, Pub/Sub, Cloud SQL, Dataproc), Azure (Databricks, ADLS Gen2, ADF), Docker, Kubernetes, Terraform, GitHub Actions CI/CD
ML, AI & APIs	LangChain, LangGraph, multi-agent/ReAct, RAG, pgvector, Amazon Titan V2, LLM evaluation, feature pipelines, model serving, FastAPI (60+ REST endpoints), Spring Boot
Data Quality & Governance	Automated DQ frameworks, schema enforcement, anomaly detection, lineage, observability, SLAs/SLIs, OpenLineage, HIPAA/HITRUST/SOC 2, PHI/PII safeguarding; Tableau, Power BI

PROFESSIONAL EXPERIENCE

Zakipoint Health — Morrisville, NC (Remote)
Senior / Lead Data Engineer — *July 2020 – July 2025*

- Architected enterprise healthcare data platform on GCP & AWS processing **10TB+ daily** across **50+ health plan clients**; designed greenfield Data Mesh with domain-oriented data products, self-service access, and federated governance using Databricks, Delta Lake, PySpark, Airflow, and dbt—scaled from 2–3 to 20+ clients.
- Built **150+ Python/FastAPI** and **100+ Java/Spring Boot microservices** with AES+RSA encryption, JWT auth, and rate limiting; achieved **99.8%+ uptime**, **sub-200ms P95** on multi-billion-row queries, and 200K+ daily API calls.
- Engineered Kafka streaming and batch ETL/ELT integrating Epic EHR via HL7 v2/FHIR R4 (ADT, ORU, 837/835, 270/271); reduced data latency **95%** (2 days → 10 min) and pipeline failures **60%+** through partitioning, caching, and shuffle tuning.
- Owned AI/ML Cost Estimator product end-to-end: built feature pipelines from claims/visit history and engineered **hybrid semantic search** (Amazon Titan V2 1024-dim, pgvector, cosine similarity + structured pre-filters) enabling natural-language provider matching across 5M+ records under HIPAA.
- Standardized IaC with **Terraform and AWS CDK** (S3 zones, Glue/EMR, Step Functions, IAM, CloudWatch)—reduced provisioning time **70%**; managed **200+ daily Airflow DAGs** at **99.8% uptime** with dbt and SLA tracking; built DQ/observability frameworks (schema enforcement, anomaly detection, OpenLineage) reducing inconsistencies **40%+**; mentored **8 engineers**, promoted **3 to senior**.

Om Hospital & Research Centre — Kathmandu, Nepal
Data Engineer / Backend Developer — *April 2018 – July 2020*

- Built organization's first clinical data warehouse on GCP (BigQuery, Cloud SQL, GCS, Dataflow) processing **250K+ annual patient visits**; designed star schema, incremental ETL, and telemedicine workflows (**10K+ monthly interactions**)—improved reporting accuracy **30%**, reduced manual workflows **70%**.
- Led zero-loss migration from paper records to DHIS2; trained **50+ clinicians and staff** on digital workflows, improving adoption and data quality organization-wide.

FEATURED PROJECTS

FFIEC Bank Analytics Platform with LangChain Multi-Agent AI [Live Demo] [GitHub]
PySpark ingestion on GitHub Actions CI/CD → Parquet on Cloudflare R2 → DuckDB httpfs engine (**sub-200ms**, **40x speedup** vs. 8–15s baseline); FastAPI SSE streaming to React/Vite; LangChain + LangGraph multi-agent assistant (master orchestrator, 2 ReAct sub-agents, 8 tools, thread-isolated memory) over 8,000+ U.S. banks.
Python, PySpark, DuckDB, FastAPI, LangChain, LangGraph, Gemini, React, GitHub Actions

McKesson Enterprise Analytics & Recommendation Platform *(W&M MSBA Capstone)* [GitHub]
Fortune 9 healthcare distributor: **110.4M transactions, $8.23B revenue**. 8-signal recommender → 3.69M recommendations; FastAPI (53 endpoints, sub-500ms P95); PostgreSQL + DuckDB; temporal holdout **91.9% vs. 70.2%**; cart simulation **49.94x lift**; **266 tests at 98.87% pass**.
Python, FastAPI, DuckDB, PostgreSQL, scikit-learn, scipy, mlxtend, React

Healthcare Patient Churn & Member-Risk Propensity Modeling *(HIPAA-Protected)*
End-to-end ML pipelines on **5M+ longitudinal member records**: feature engineering across claims, enrollment, and demographics; logistic regression, random forest, XGBoost with ROC-AUC and cross-validation; deployed scoring pipeline integrated with retention and care management workflows.
Python, R, SQL, scikit-learn, XGBoost, pandas, statsmodels

EDUCATION & CERTIFICATIONS

William & Mary, Mason School of Business – **Masters of Science in Business Analytics** — August 2025 – May 2026
Vellore Institute of Technology – **Bachelors of Technology in Computer Science Engineering** — August 2018 – May 2022
Certifications: Azure Databricks for Data Engineers (Packt/Coursera) • Data Analysis with R (Duke/Coursera) • Google Data Analytics (Google/Coursera)